

Systematic approaches to network infrastructure planning and performance modelling

J Mellis

Developments in database technology, software design, and operational research methods allow powerful new systematic approaches to modelling and planning the infrastructure of telecommunications networks. These developments are timely — they accompany a trend towards increased technical complexity and diversity of infrastructure plans (as evidenced by the emerging hybrid fibre/copper/radio access network). They also assist in meeting the rediscovered need to understand the detailed interactions of plant and people, consistent with the imperatives of business process re-engineering.

1. Introduction

The traditional complexity and diversity of telecommunications network performance engineering requires the fragmentation of network performance problems into various sub-disciplines, such as transmission design, capacity planning, reliability engineering, ergonomics and cost modelling. In most cases a single network will be represented in tens or hundreds of different data models, for the multifarious purposes of design, planning, management and operations, and this situation is not likely to change quickly. However, the increasing availability of cheap, distributed computing power, together with recent developments in relational and object oriented databases, offers new possibilities for a more holistic approach to network performance modelling. The most fundamental aspects of network performance (transmission performance, reliability, and cost) can now feasibly be considered simultaneously and interactively.

This paper reviews these possibilities as they relate to the planning and performance of the network physical infrastructure layer (cables, cable nodes, and other external plant), and refers to the development of a number of models and tools which, taken together, provide the beginnings of a comprehensive description of the infrastructure performance. The physical infrastructure layer is a good place to begin the performance planning and modelling task. Comprehensive models of the network infrastructure performance can be built using a relatively small set of network elements; and, as the ultimate service-independent network platform, the cable infrastructure's performance forms the basis of all other network quality of service measures. Figure 1 illustrates the whole performance

domain of interest, and gives some examples of relevant engineering and planning considerations.

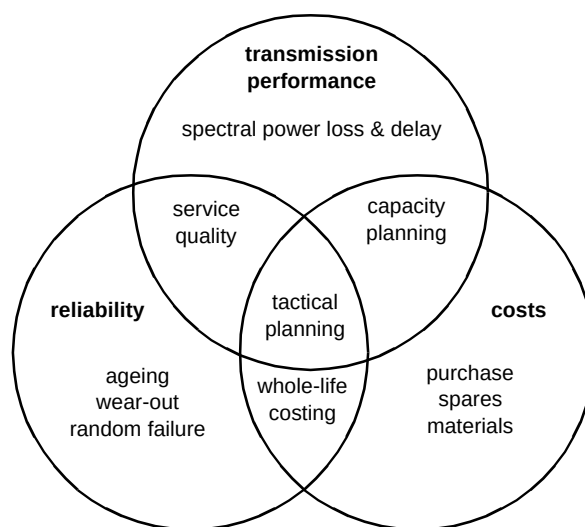


Fig 1 Holistic view of some network infrastructure performance issues, and associated planning considerations.

2. Infrastructure transmission performance

Perhaps the most basic telecommunications network performance requirement is the transmission performance of the bearer (i.e. copper pair, optical fibre, coaxial cable, etc). Much new interest has been stimulated in the

high-frequency performance of unshielded twisted pair (UTP) cables by the emergence of digital transmission systems such as asymmetrical digital subscriber line (ADSL), offering 2–6 Mbit/s downstream data rates over up to 6 km of twisted-pair cable, and its very high-rate equivalent (VDSL) which offers the prospect of broadband multimedia delivery to the home over standard telephone lines [1].

The frequency-dependent transmission loss of twisted-pair cable is principally a function of cable length, pair gauge, insulator material and cross-sectional geometry. Models accurately describing the frequency dependence of loss are well developed, but the prediction of crosstalk between pairs and cables is less well understood, and crosstalk and radio-frequency interference (RFI) are likely to be the limiting factors for very high-rate transmission. New models [2] which describe the near- and far-end crosstalk (NEXT, FEXT) and the RFI performance of UTP cables are now being tested against the anticipated stringent operating requirements. The modelled transmission capacity of standard unshielded twisted-pair telephone cable is a complex function of cable length, pair gauge, number of pairs in the cable and the nature of the insulator material, as well as the digital systems 'fill' of any given cable, which determines the number of crosstalk interference sources. Modelling the transmission capacity of a typical line against all these constraints is challenging, but can be achieved [3] by making appropriate generalising assumptions. Figure 2 illustrates the projected transmission capacity of a 'typical' telephone line, as a function of cable length, for ADSL and VDSL systems currently in development.

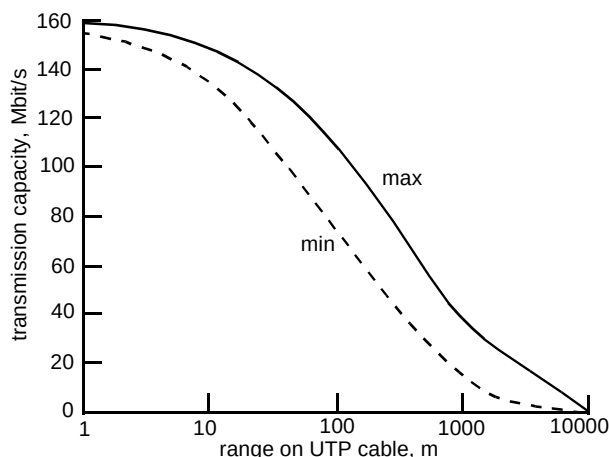


Fig 2 Transmission capacity of conventional unshielded twisted-pair (UTP) telephone cables as a function of cable length, based on simulated use of ADSL and VDSL transmission systems (after Kirkby [3]).

Empirical field surveys to test these transmission capacity modelling assumptions are required, and are being undertaken by BT and other companies [4]. Experience

suggests that significant variation between expected and measured line transmission performance can occur, influenced not only by the cable length and make-up factors already described, but also by the variable adaptive capabilities of the new transmission systems and the accuracy of distribution network external plant records. Understanding the nature and extent of these performance variations is crucial to the development of deployable ADSL/VDSL systems, and will also help to improve the tactical planning of present-day digital systems such as pair-gain multiplexing and ISDN systems.

Figure 3 illustrates schematically how uncertainty in the estimation of copper line transmission performance leads to inefficient planning. The shaded area represents the range of variation between expected and actual line losses for a given hypothetical digital transmission system with a loss budget of 55 dB at its equivalent centre frequency. The extent to which the targeted lines populate the shaded area determines the planning efficiency for the service in question. Lines falling in the lower left quadrant of the diagram are 'in range' as projected from plant records and transmission loss models. Lines falling in the top right quadrant are likewise 'out of range' as modelled. However, lines in the second or fourth quadrants cause difficulty, leading either to 'lost business' (where the line is calculated to be outside the required specification, but is in fact usable) or to planning errors and service provision failures (where a line proves unusable despite a positive performance estimation in the planning phase).

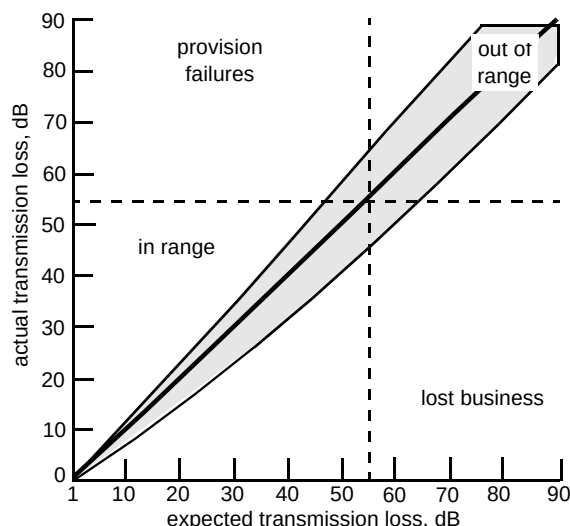


Fig 3 The effects of variation in network line plant performance on the efficiency of service provision planning.

Even present-day narrowband digital systems can suffer from such planning uncertainty, often leading to time-consuming and costly line tests to decide the feasibility of

specific deployments. The value of accurate electronic plant records, and good transmission performance models, could hardly be better illustrated than in this area of detailed network planning.

Similar requirements for accurate transmission loss estimation apply to the planning and deployment of fibre infrastructures, especially in the local network, where point-to-multipoint passive optical networks (PONs) are being rolled out. This splitter-based topology efficiently shares costs between customers and maximises the exploitation of valuable exchange-side duct space. The penalty is some complication of network planning and management. PON transmission losses are determined by the combination of splitter, cable and splice losses in a way which is best represented by a statistical loss model, which specifies the mean and variance of losses imposed by all the passive optical components [5]. This should take account of the variability of loss which is observed from component to component, and between ports of individual splitters, offering a more accurate result than the simplistic, conservative loss estimates which have usually been applied to point-to-point fibre link designs in the past, often leading to the use of attenuators to bring the final link-loss into specification. A statistical approach to the specification of optical local network power losses has been adopted by the European Telecommunications Standards Institute [6] and the paper by Bell et al [7] illustrates well how this approach has been implemented in a power-budget estimating tool which is now in use by BT network designers.

An even more accurate model of optical network transmission loss is possible if the spectral attenuation and scattering-effects of the optical fibre are also included. In this case, detailed spectral loss simulations can be performed, not only for forward losses, but also for return losses, as determined both by Fresnel reflections and Rayleigh scattering. This opens the intriguing possibility of simulating the network's response to optical time or frequency domain reflectometry (OTDR, OFDR), with many potential applications in network planning and management. The paper by Asumu et al [8] describes the design, construction, and testing of such a spectral modelling tool, and its possible uses in network design, performance monitoring systems, and fault diagnostics.

3. Infrastructure cost modelling and optimisation

An increasingly detailed understanding of network capital and operating cost is important because of growing competition in telecommunications markets, rapidly diversifying options for network technology, and, at least in the UK, ever-tightening regulation and demand for accounting separability.

Even for advanced new optical networks, the capital costs are often dominated not by transmission equipment

but by cable, installation, and civil engineering costs [9, 10]. A consistent treatment of these cost elements is also required for many business-as-usual activities such as construction, repair, diversion or rearrangement of copper pairs, external plant and street furniture. At the most basic level, improved tools are required for the quick-but-consistent estimation of installed costs of cables, street cabinets, footway boxes and poles. This is true both for new network schemes, and diversionary works applied to existing plant.

A budget estimate calculating system (BECS) has recently been developed by Butler [11] which produces rapid budgetary estimates for external plant diversionary works. These can be forwarded as client estimates, or used as the basis for more detailed cost analysis. The system, written in Visual Basic, is menu-driven (see Fig 4), simple to use, and contains a cost model based on a controlled, upgradable file of standard costs for stores, materials, direct labour and external contracts. BECS is currently being deployed in BT planning offices, and it provides an excellent foundation for a wider range of standardised network infrastructure capital cost models. These could apply alike to new developments on green-field sites, or future network upgrades to accommodate fibre-to-the-cabinet for broadband services.

A more specialised costing problem occurs when we consider passive optical network (PON) planning options, and the allocation of costs across customers connected or passed by the network. In this case cost modelling is essential to identify the optimum network dimensions, specifically with respect to cable size (fibre count), splitter sizes, and splitter node location. The cost-modelling overview by Bell et al [12] provides example results from such cost optimisation, and describes their use in the development of network planning policy and tactical planning rules.

Decision-making using rules (like those contained in BT Core Planning Guides) is a process very suitable for automation through computer-based expert systems. Progress in artificial intelligence and object oriented programming [13] is already producing commercially available knowledge-based systems development tools, such as G2, which facilitate schematic reasoning [14]. These can be used to create PC-based, natural-language or graphics-driven dialogues that guide human users efficiently through complex decision trees. Similarly, the growing power of distributed computing means that complex planning problems can be segmented for attack by solver modules that address particular aspects of the optimisation problem, but which still use a shared core of performance data. For example, choosing cable routes to maximise the exploitation of available duct space, or selecting positions and sizes of customer access nodes to

BECS - Duct / External Plant

Demolish Manhole		Manhole Build		Jointbox Build	
☐ 1	☐ 1 MRX404	☐ 1 MRX411C	☐ 0 None	☐ 0 None	☐ 0 None
Demolish Jointbox		Jointbox Build			
☐ 0	☐ 2 JUF10	☐ 0 Unknown	☐ 0 Unknown	☐ 0 Unknown	☐ 0 Unknown

Duct (500mm cover)				Duct Extras		
Route	Number	Length	Soft F/W C/W		Rate £	Total £
1	4	58	C G C	Frame+Cover FW (W/O Stores)	187.52	375.04

BECS - Copper D/E-Side Cables

D-Side Cables							
Index	Description	Pairs	Length	Ends in MH	Int. Joints	IJs in MH	Cable Recovery
1	Evesham Road (N-S)	50	134	0	0	0	Yes
2							
3							
4							
5							
6							
7							
8							
9							

D-SIDE

Number	Customer connections	Poles Block Terminals
☐ 2	☐ 5	☐ 2

Attachments	Pairs	Aerial Cable Length
☐ 1	☐ 5	☐ 15

Cabinet Internal Wiring

D/S pairs ☐ 0 E/S pairs ☐ 0

D-Side Cable Details

SELECT D-SIDE PAIR COUNT

Length (m)

Cable Recovery? ☒ Yes

Cable Ends in Manhole ☒ 0 ☐ 1 ☐ 2

Intermediate Joints

Total Number

Number in Manhole

Cable Description

E-SIDE

Grid Instructions

CTION LABEL for cable type (e.g D-Side)

GREY INDEX column to activate grid

delete selected row upon data entry

Sub-totals	
0.03	UCA 3.00
0.00	UJ 32.73
0.36	STORES £ 746.40
0.00	
0.13	

Fig 4 Screens from BECS showing menu-driven approach to standardised external plant cost estimation.

minimise cable and plant costs, are problems which are increasingly amenable to computerised solutions. The papers by Fisher et al [15] and Paul and Tindle [16] demonstrate the application of advanced optimisation algorithms to such practical problems. Ultimately, the use of optimisation techniques may allow the elimination of planning rules completely, leaving only the loosest constraints to the auto-optimisation problem.

To end the discussion of cost models, the emphasis must be moved from capital (non-recurring) to current (recurring) network costs. The estimation of costs and cash-flows over the whole life of plant and systems is already the major determinant of BT procurement decisions, driven by the use of 'net present value' methods in capital investment decision-making [17], and the use of tools such as the whole-life investment appraisal (WINA) model [18]. Unfortunately, getting realistic answers from whole-life costing techniques often requires more knowledge of equipment reliability, maintenance burdens, and repair costs

than can easily be obtained. For this (and other) reasons, reliability models form a crucial part of the overall plant performance view, as emphasised in the next section.

4. Reliability — plant, processes, and people

Any systematic view of network reliability usually begins with the identification of appropriate network elements and the definition of real or judged values for such parameters as mean time between failures (MTBF), and mean time to repair (MTTR) for these elements. By these standards the reliability of telephone networks is high (with single line MTBFs of greater than 5 years in the UK and USA), and, maybe more importantly, telephone company customers generally reinforce this perception of high-quality service.

At the detailed level of the cable infrastructure, great caution must be applied to the use of means and medians to describe the customer perception of network reliability, especially where the access network architecture implies that a large minority of customers can experience poor

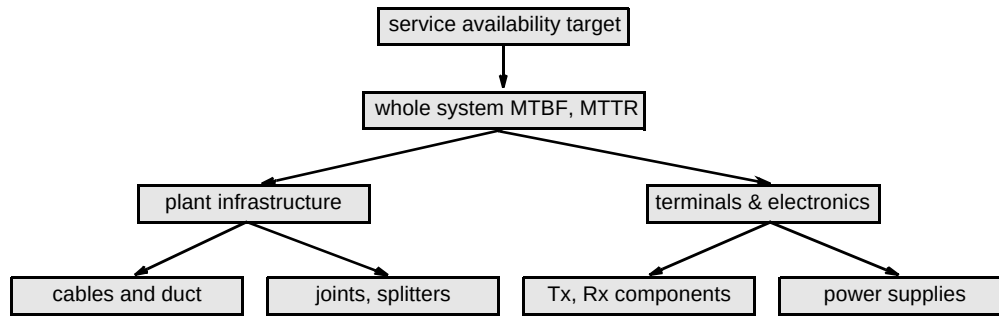


Fig 5 A 'top-down' approach to the identification of reliability targets for network components

reliability levels 'in the tail of the distribution'. For example, historically, cable TV networks have provided a less reliable service [19] for some customers whose service is compromised by the use of many externally-located line amplifiers, arranged in cascade, between the coax distribution node and the customer premises. Nevertheless the quotation of mean levels of service reliability and outage time is the standard method of defining network quality of service. More recently, service availability targets have also been used as the starting point for the specification of sub-system and component reliability requirements through the 'top-down' apportionment of the total reliability budget, in a systematic way, between all the elements of the network infrastructure including cable and passive components [5]. Figure 5 illustrates this process, which when completed allows the reliability targets for network components to be balanced, and optimised against the other important considerations such as cost and human factors.

It is interesting to review the behaviour of BT's network fault rates over the last few years. When the volume of network faults is normalised against the growing network size, the fault-rate (in terms of faults per exchange-line per year) shows clear improvement, driven largely by improved network switching, transmission and outside plant technology. The overall volume of faults (and hence the time and costs of repair work) have remained fairly static. A generally high level of reliability is provided by the BT public switched telephone network, with an MTBF of more than 5.5 years (failure rate better than 0.16 per line per year). For private circuits, the increased complexity of line plant and customer premises equipment implies a higher level of failures of around 0.5 per circuit per year. Again this has improved steadily in the last few years, as have most other quality-of-service indicators such as the fraction of business customer faults repaired within the target time (Fig 6). On a more detailed level (e.g. month on month, or for specific geographical areas) a much more dynamic picture emerges, especially for the external network, which is subject to the vagaries of weather, fluctuating levels of repair and maintenance activity, cable dig-ups, and wide variations in plant age and condition. An understanding of

the short-term dynamics of the external network fault rate is an essential step to quantifying the underlying reliability and condition of the outside plant. The paper by Thomas et al [21] describes a mathematical model (FRAME) which accurately characterises the dynamic behaviour of the external network fault rate in terms of only three basic input parameters:

- a background level of (random) plant failures,
- weather-induced failures (related to rainfall and wind speed),
- manpower-induced faults (related to measurable activities such as network construction, pair provision and installation, and fault location and repair).

That such a simple model works so well is encouraging, and offers prospects for some predictive capability for network fault rates, more meaningful comparisons between different time periods or geographical zones, and new network performance indicators to measure not only levels of plant condition, but also the quality of manual network intervention.

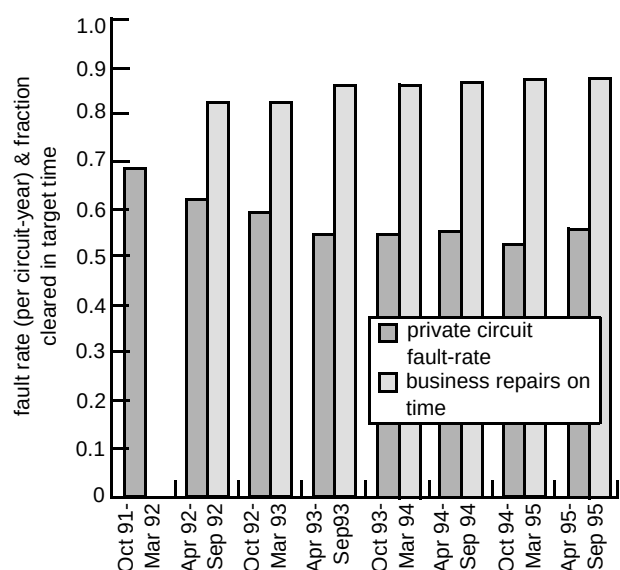


Fig 6 Two BT network quality of service indicators 1991—95 [20].

Here a truism is encountered that is easily overlooked — network performance at all levels is fundamentally affected by human factors. The speed and quality of maintenance, fault location, defect repair, planning decisions, and installation practices are critical influences, particularly for the external network, where these process issues may dominate the ‘designed’ performance of the plant. Figure 7 illustrates some performance-related

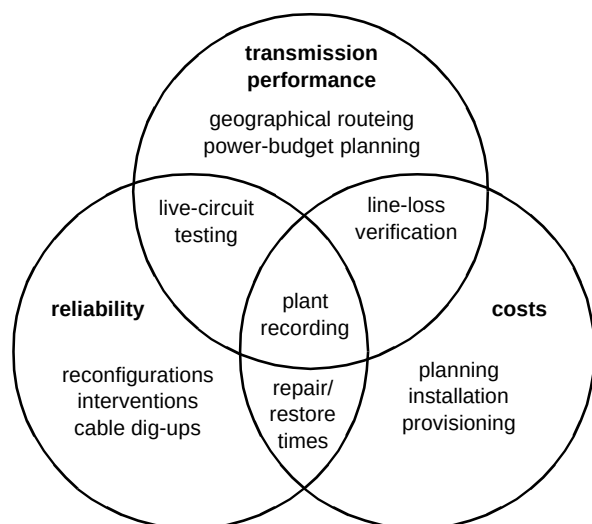


Fig 7 Some performance-critical processes and activities.

processes of importance, and positions them within the transmission/reliability/cost triangle. Repair times and repair quality are obviously crucial influences both on network running costs, and, in the long run, on network reliability. Transient transmission errors can occur on single-mode optical-fibre circuits [22] as a result of network reconfiguration or fibre provision activity — carefully designed external fibre plant, such as BT’s OTIAN™ system [23], is required to eliminate these effects. Fault diagnosis, whether manual or automated, is always intrusive to some degree, and also poses at least a transient risk to the integrity of the transmission network. Provision planning uncertainty, as already shown, can require costly line tests to verify specifications — and clearly the development and maintenance of accurate, electronically accessible external plant records is central to improving the quality of all these activities.

A commonly shared, readily accessible cache of plant performance information has become a real possibility in the last few years, through the development of relational and object oriented (OO) databases. In particular the OO characteristics of hierarchy, inheritance and polymorphism [13] are ideally suited to the efficient modelling of telecommunications networks. Data objects can now

naturally mirror real plant objects, and encapsulate all their performance characteristics in a way which allows them to be easily and consistently assembled into a huge variety of network models. The paper by Redstall et al [24] describes the construction of an object-based model of the network infrastructure (OMNI), used initially to model plant reliability, but intended eventually to provide a universally applicable database of plant component cost, performance and reliability information. Such developments could eventually lead to the fusion of disparate network models into global, connected, end-to-end performance models as already envisaged by far-sighted commentators [25].

5. Conclusions — towards total network performance simulation and planning

The developments already reviewed are beginning to realise the prospect of a set of network performance models which are:

- modular,
- distributed,
- interconnected,
- based on shared data,
- easy to access and use,
- holistic (i.e. forming a system covering transmission capacity, reliability, and cost parameters).

Further incorporation of the process-related performance issues, especially the effects of human activity and intervention, would allow a total picture of the network performance to be developed (Fig 8). In such a total model, the effects of hardware component design changes on, say, end-to-end circuit availability, repair times, operational costs, and maintenance resources could be speedily estimated. Recent developments in the more precise definition of processes (driven by the need for process re-engineering) clearly assist in the construction of such a comprehensive vision. Tindle et al [26] enlarge upon this topic, and review some promising approaches to total network simulation, for the purposes of network management and business modelling, as well as the more near-term prospects for advanced planning systems.

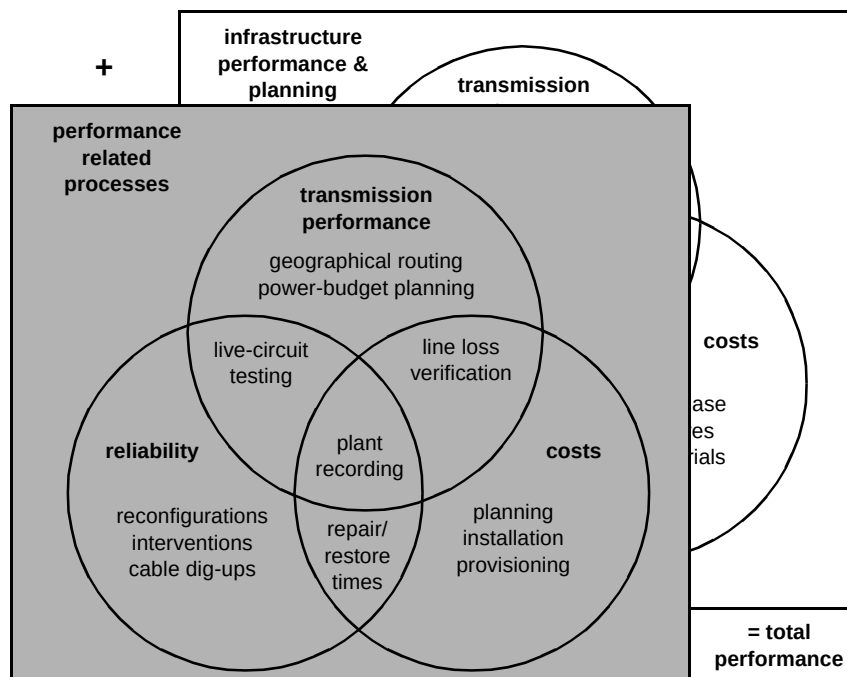


Fig 8 The combination of hardware performance with related processes to produce a view of total network performance.

Of course, an enormous amount of work needs to be done to realise even some of these possibilities. Today, the vast proportion of external plant records exists only in the form of paper maps, although BT, in common with most telcos and utilities, has embarked upon scanning, digitisation and data-cleanse projects. The prospects for accelerating these projects are good, thanks to rapid advances in the areas of image and symbol recognition, raster-to-vector conversion, and other progress [27] within the generally buoyant field which has become known as 'automated mapping, facilities management, and geographical information systems' (AM/FM/GIS). This closely related but distinct subject therefore has great and growing importance for external network planning. However, in the interests of minimising space and maximising focus, its exploration must be left for future publication.

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John Mellis leads the Network Engineering Plant and Pilots unit at BT Laboratories, which specialises in the performance, planning, and pilot trial of network infrastructure components, cables and systems. He joined BT in 1986 to engineer network demonstrations of coherent optical transmission, and from 1989-91 was with BT&D (now HP Optoelectronics), commercialising optical amplifier and tunable laser products. From 1983-86 he was at STL, Harlow (BNR Europe), developing electro-optic/acousto-optic signal processors. He received a doctorate from the University of St Andrews in 1983, for the mathematical modelling of CO₂ laser kinetics, and graduated first-class in Applied Physics (with Operational Research) from the University of Strathclyde in 1978. He is chairman of the ETSI expert group on passive optical components, sits on the technical advisory committee of the European Conference on Networks and Optical Communications, and is a Member of the IEEE.